



Quantitative analysis of diffusional pore coupling from T_2 -store- T_2 NMR experiments

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ABSTRACT

Low field NMR T_2 distribution is an efficient tool for characterizing pore size distribution in porous media. In NMR relaxation experiments, we measure the magnetization decay characterized by the so-called T_2 relaxation time, resulting essentially from liquid–solid interactions of the spins carried by molecules and exploring the pore space by diffusion. For multimodal systems, a new technique called T_2 -store- T_2 allows the analysis of diffusional pore to pore exchange that is extremely useful for the characterization of connectivity. This technique uses 2D inverse Laplace techniques and produces T_2 – T_2 maps. Qualitatively, a system is coupled when off-diagonal peaks are observed.

Based on an analytical solution describing diffusional coupling between two pore populations, we propose defining a coupling factor that quantify the degree of coupling between the two populations, providing an easy understanding of the complex analytical solution. This analysis allows understanding 1D T_2 experiments as well, and indicates some limitations of the T_2 characterization when interpreted as a pore size distribution.

We provide examples of bimodal pore structures in which we applied our methodology: a clay gel system, a shaly sandstone and two double porosity carbonates. These systems are also described by conventional techniques (mercury injection, SEM visualization) and illustrate weak, intermediate and strong coupling. Despite the presence of distribution of pore sizes, the two pore system exchange model gives satisfactory results for the quantitative analysis of the coupling and T_2 -store- T_2 experiments. In carbonates, more complex exchanges can occur between micro-, meso- and macropores.

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1. Introduction

Low field NMR relaxation is a unique non-invasive and non-destructive technique to characterize pore size distribution of porous media. Spin bearing molecules explore the pore space by diffusion and provide a unique information about the volume to surface ratio of a given compartment. This principle was first evidenced by Brownstein and Tarr [1] in biological cells and relies on a contrast between surface and bulk relaxivity. It is used extensively for the interpretation of NMR relaxation data in natural porous media especially in the field of petrophysics. Without considering the additional difficulty of the fast and slow diffusion regimes of relaxation [2], i.e. we consider typical pore sizes and surface relaxivity values, the pore size interpretation is thought to be usually valid because the diffusion length during the lifetime of the magnetization is of the same order of the distance between grains. Hence, the exchange of molecules between pores should be limited. However, there are situations where this is not the case.

Diffusional pore coupling is an issue of practical importance, especially in the field of logging interpretation. NMR relaxation data should not depend on temperature and this fact was put forward to justify the surface dominated regime [3]. However, in carbonates, a significant temperature dependence can occur, typically when comparing laboratory and log data performed, respectively, at 20 and 100 °C. This is a direct effect of pore coupling and a solution to predict the shift of the T_2 distributions was proposed [4]. At the extreme situation, a multimodal system may appear unimodal and this has important interpretation consequences since the NMR log is the only in situ measurement sensitive to the pore structure. Later, Anand et al. [5,6] proposed a 1D geometric model and a coupling parameter to describe the pore to pore coupling phenomenon and applied it to various situations. For bimodal pore systems, the shift of macro- and micropore NMR peaks can be predicted as a function of the coupling parameter comparing the characteristic relaxation rate to the diffusion rate.

From the NMR point of view, the methods for observing the modification of the environment of spin carrying molecules has been evidenced very early in the development of NMR [7]. At high field, the analysis of the chemical exchange is now a common practice, for example in the exchange spectroscopy technique EXSY [8]

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that uses very similar principle as used here. The results of these experiments is a 2D map of chemical shift in which off-diagonal peaks signify a change of chemical environment after a chosen exchange time. At low field, the observation of the modification of the spin carrying molecules in terms of relaxation is recent. Simultaneously, two groups [9,10] proposed new NMR sequences to detect the relaxation exchange of molecules in the case of a cement paste and a sandstone. The experiment consists of two T_2 relaxation periods with a variable exchange time sandwiched between them. When analyzing the resulting 2D T_2 - T_2 maps, the pore coupling is evidenced by off-diagonal peaks. The motions of molecules can be observed only if a surface relaxivity contrast exists between T_1 and T_2 [9,11] which is usually the case in porous media. More generally, the existence of off-diagonal peaks can also be analyzed from the resolution of eigenvalues of the underlying Bloch-Torrey diffusion equations and some general behavior can be deduced [11].

In this work, we first analyze the analytical solution proposed by Montheillet et al. [9] and propose a coupling parameter based uniquely on NMR measurable factors to quantify the degree of coupling between two pore sizes. Then, the method is applied to different systems with a large range of sizes, from nanometers to tens of microns. We study the pore to pore exchange in a natural bimodal model system composed of flocculated clay particles, and samples typical of oil bearing rocks: a high permeability sandstone containing clays and two complex carbonates.

2. Theory: coupling parameter

We first consider the case of complete coupling using a simple fast exchange model. Consider two porous zones in a porous media (Fig. 1), typically representative of micro- (subscript a) and macropores (subscript b). For simplicity, we will assume a unique pore size in each zone. In these conditions, the NMR relaxation time in the surface dominated regime in the absence of coupling will be proportional to the volume to surface ratio of the macropores $T_{2b} \propto V_b/S_b$ and micropores $T_{2a} \propto V_a/S_a$. The measured distribution will be bimodal. In this case, diffusive molecular motions are such that the magnetization has decreased to zero before the molecules can travel to a different porosity zone.

In the case of complete coupling, molecules explore both environments during the typical life time of the magnetization. Assuming that relaxation is still in the fast diffusion regime, the observed relaxation time T_2^+ (the notation is linked to the theory described later) is then proportional to the total volume to surface ratio according to:

$$T_2^+ \propto \frac{V_T}{S_T} = \frac{V_a + V_b}{S_a + S_b} \quad (1)$$

Each volume V_a and V_b corresponds to a magnetization M_a^0 and M_b^0 . Assuming a unique surface relaxivity for both a and b zones, the observed relaxation time T_2^+ is unique and smaller than the macropore peak T_{2b} according to:

$$\frac{T_2^+}{T_{2b}} = \frac{M_a^0 + M_b^0}{T_{2b} M_a^0 + M_b^0} \quad (2)$$

For example, when the pore sizes are different by a factor of 10 ($T_{2b}/T_{2a} = 10$), the observed single relaxation time $T_2^+ = 0.2T_{2b}$ when $M_a^0 = M_b^0$. The relaxation time T_2^+ is closer to the micro-pore peak than the macro-pore because there is much more surface in the microporosity.

If the coupling is intermediate, we now use the theory developed by Monteillet et al. [9] describing the full details of the diffusional coupling between two zones a and b , and we show that a single parameter can be build to characterize the degree of coupling. For spin-spin relaxation, the system of coupled differential equations is given by:

$$\begin{aligned} \frac{dM_a}{dt} &= -k_a M_a + k_b M_b - R_{2a}^0 M_a \\ \frac{dM_b}{dt} &= -k_b M_b + k_a M_a - R_{2b}^0 M_b \end{aligned} \quad (3)$$

where $k_{a,b}$ is the exchange rate, $R_{2a,b}^0$ is the nuclear spin-spin relaxation rate ($= 1/T_{2a,b}^0$) of each zone in the absence of coupling. At $t = 0$, the magnetization in each reservoir is $M_{a,b}^0$; the exchange rates $k_{a,b}$ are linked by the balance equation:

$$k_a M_a^0 = k_b M_b^0 \quad (4)$$

The system of Eq. (3) was solved analytically [9] in the specific context of the T_2 -store- T_2 sequence detailed in Appendix A. Essentially, the relaxation is first observed using a CPMG sequence of variable length. Then, the magnetization is stored along the z -axis (T_1 relaxation) to allow spin bearing molecules to move by diffusion in a different environment (a different pore system). Also, because the surface relaxivity is slightly different, the relaxation decay during the storage or exchange period is different. Finally, the magnetization is returned to the x - y plane and the decay recorded. The result of the experiment is a T_2 - T_2 map containing four peaks in the case of coupling (Fig. 2). The observed relaxation rates $s_{1,2}^{+/-}$ ($= 1/T_{1,2}^{+/-}$) are given by:

$$\begin{aligned} s_{1,2}^{+/-} &= -\frac{1}{2} (R_{1,2}^a + k_a + R_{1,2}^b + k_b) \\ &\pm \frac{1}{2} \sqrt{(R_{1,2}^a + k_a + R_{1,2}^b + k_b)^2 - 4 \{ (R_{1,2}^a + k_a)(R_{1,2}^b + k_b) - k_a k_b \}} \end{aligned} \quad (5)$$

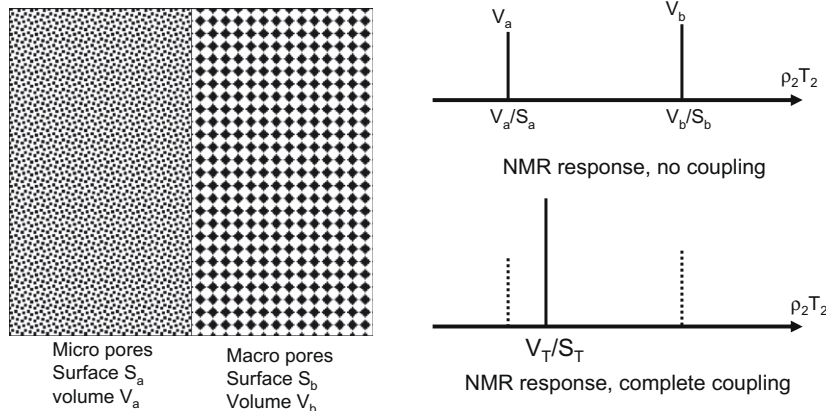


Fig. 1. Illustration of the pore coupling effect on 1D T_2 relaxation time distributions. The pore structure information may be lost; micro- and macropores can no longer be identified in the case of complete coupling.

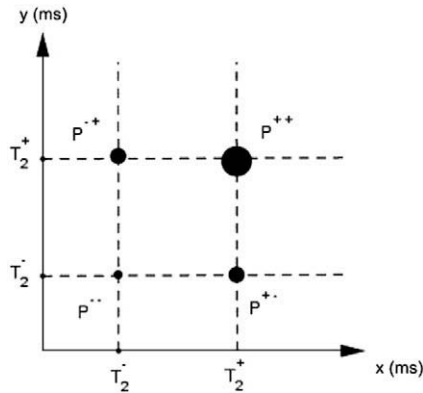


Fig. 2. Illustration of the positions and amplitudes of diagonal and off-diagonal peaks as a result of diffusional coupling between two reservoirs, as measured in a T_2 -store- T_2 experiment.

The observed amplitudes of the diagonal peaks P^{++} and P^{--} , and off-diagonal peaks P^{+-} and P^{-+} , are function of the exchange time t_e that can be varied experimentally:

$$\begin{aligned} P^{++} &= C_1 \exp(s_1^+ t_e) + C_2 \exp(s_1^- t_e) \\ P^{--} &= C_3 \exp(s_1^+ t_e) + C_4 \exp(s_1^- t_e) \\ P^{+-} &= P^{-+} = C_5 \exp(s_1^+ t_e) + C_6 \exp(s_1^- t_e) \end{aligned} \quad (6)$$

where the coefficients from C_1 to C_6 are complicated functions of $s_{1,2}^{\pm}$, $M_{a,b}^0$, $R_{1,2}^{a,b}$ and $k_{a,b}$ (see Appendix B). Note that the ratio T_1/T_2 is included in these coefficients. These peaks can be observed experimentally using 2D inverse Laplace techniques described later. The above model can be used to show the shift of the micro- and macropore peaks in a 1D NMR CPMG relaxation experiment, and to explore the behavior of 2D experiments for different degrees of coupling.

Due to the complicated solution of Eq. (3), it is rather difficult to analyze the effect of the numerous parameters included in the system. For this purpose, we suggest to use the following coupling parameter:

$$\alpha = \frac{(\text{average exchange rate})}{(\text{micropore relaxation rate})} = 2(k_a + k_b)T_{2a} \quad (7)$$

There are other possibilities for defining a coupling parameter. Here, we base our parameter on the micropore relaxation rate (or the smallest pores) because microporosity is a major source of relaxation due the much larger amount of surface compared to macroporosity. We will also see that the use of this parameter greatly simplifies the analysis of the system. Note that α is directly proportional to the diffusivity of the saturating liquid because we expect a direct dependence of exchange rate with diffusivity. Using Eqs. (4)–(6), we calculated four quantities of practical interest by varying the exchange rate k_a by several orders of magnitude; random choices of the pore size ratio T_{2b}/T_{2a} and magnetization ratio M_a^0/M_b^0 were selected in the intervals [5,50] and [0.05,1], respectively. This means that we consider situations where the micro- and macropores are well separated and microporosity does not exceeds macroporosity. We used $T_1/T_2 = 1.5$, a typical value for sandstones and carbonates at a Larmor frequency of 2 MHz.

The four quantities of interest are plotted in Fig. 3:

- The ratio P^{--}/M_a^0 calculated at $t_e = 0$ represents the observed microporosity amplitude relative to the true one in a 1D experiment. It mostly follows a S-shaped curve that has been found using a very different approach [5]. To go from a weak coupling ($\alpha < 0.1$) to a strong coupling ($\alpha > 10$), the parameter α must vary by two orders of magnitude.

- The ratio $2P^{+-}/P_{\text{tot}}$ calculated at $t_e = T_{2b}$ represents the amplitude of the relative off-diagonal peak intensity in a 2D experiment. With the choice $T_1/T_2 = 1.5$, it has a maximum at $\alpha \cong 1$; this maximum will occur at smaller α when increasing the ratio T_1/T_2 (at the extreme value of 4, the maximum occurs at 0.6). For strongly coupled systems ($\alpha > 10$), it means that the 2D T_2 -store- T_2 experiment is unable to evidence the coupling. One must lower the value of α using a fluid of lower diffusivity.
- Concerning relaxation times, the ratio T_2^+/T_{2b} expresses the deviation of the macropore relaxation time relative to its true value without coupling. It depends both on α and the magnetization and relaxation time ratio M_a^0/M_b^0 and T_{2b}/T_{2a} , respectively. We also verified that, for large α :

$$\frac{T_2^+(\alpha \rightarrow \infty)}{T_{2b}} = \frac{M_a^0 + M_b^0}{\frac{T_{2b}}{T_{2a}} M_a^0 + M_b^0} \quad (8)$$

which corresponds to the value given in Eq. (2) using the simple volume to surface model explained above.

- Finally, the ratio T_2^-/T_{2a} expresses the shift of the micropore relaxation time relative to its value without coupling. Surprisingly, the micropore peak is shifted to smaller T_2 values (Fig. 3) as the coupling coefficient increases, whereas one would have intuitively expected a gradual merging of the micro- and macropore peaks. This behavior has been clearly observed experimentally in model systems and predicted [5].

3. Experimental methods

We describe here some important aspects when performing 2D inverse Laplace transform (ILT) and T_2 -store- T_2 experiments.

From the theory described above, we can extract the amplitudes of the off-diagonal peaks to test the feasibility of the method in various configuration, and gather important information on how to design the experiments. Using the same random fluctuations of the ratio T_{2b}/T_{2a} and M_a^0/M_b^0 as previously, we plotted the relative off-diagonal peak amplitudes $2P^{+-}/P_{\text{tot}}$ ($P_{\text{tot}} = (P^{++} + P^{--} + 2P^{+-})$) as a function of the storage time t_e (that can be varied experimentally) for a fixed value of the coupling parameter $\alpha = 1$ (Fig. 4). It is seen that the total off-diagonal peak relative amplitude does not exceed 10% for a typical value of $T_1/T_2 = 1.5$, and this has important implications for data acquisition and processing. The graph also shows that it is sufficient to observe the off-diagonal peaks for storage time t_e at maximum up to the observed dominant peak T_{2+} . If no off-diagonal peak is observed at $t_e = T_{2+}$, it means that the bimodal system is either not coupled, or totally coupled. To describe the coupling, four independent parameters need to be calculated: k_a , T_{2b} , T_{2a} and M_a^0 . The other parameters k_b and M_b^0 are deduced from Eq. (4) and by normalizing the data set by $M_a^0 + M_b^0$. For this back calculation, we suggest to use in addition the diagonal peak amplitudes that also depends on the storage time t_e (Fig. 4). For the present conditions, the normalized variation of the macro- and micropore peaks ($P_n^{++} = P^{++}/P^{++}(t_e = 0)$ and $P_n^{--} = P^{--}/P^{--}(t_e = 0)$) are mostly described by a single exponential decay.

3.1. Off-diagonal peak detection

It is well known that ILT must be used carefully [12] to avoid spurious peak and over interpretation of the results. For 2D calculations, it is obviously more crucial. Based on efficient computing algorithms [13], we build our own software and made extensive tests to evaluate possible short comings. Essentially, one must find a matrix of relaxation times A fitting the data, such as to minimize the cost function SE:

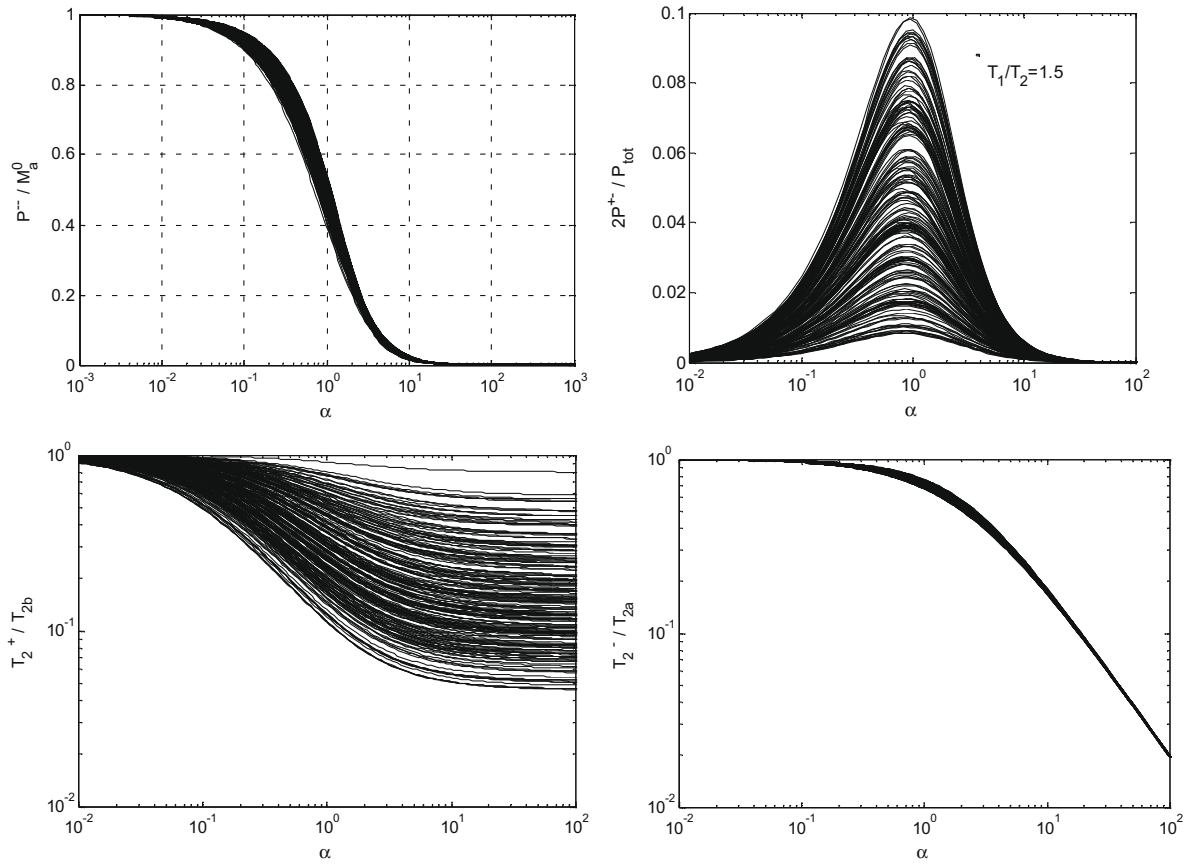


Fig. 3. Study of four quantities of interest as a function of the coupling parameter α when the ratio T_{2b}/T_{2a} varies from 5 up to 50, and M_a^0/M_b^0 from 0.05 up to 1 ($T_1/T_2 = 1.5$). Random combinations of these two ratios were chosen to calculate the curves. P^{+-}/M_a^0 : amplitude ratio of the observed microporosity peak to the true amplitude at $t_e = 0$; $2P^{+-}/P_{tot}$: amplitude ratio of the off-diagonal peak to the macroporosity peak at $t_e = T_{2b}$; T_2^{+}/T_{2b} : deviation of the macropore relaxation time relative to its true value; T_2^{-}/T_{2a} : deviation of the micropore relaxation time relative to its true value.

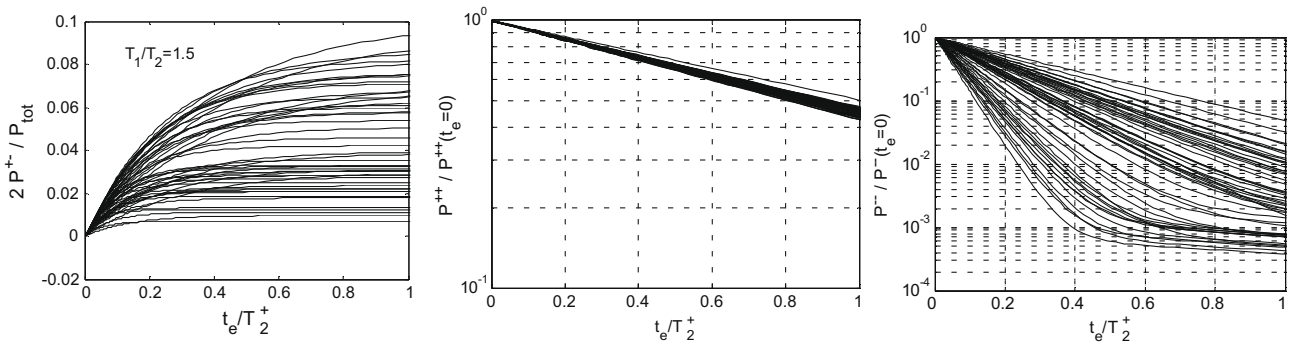


Fig. 4. On- and off-diagonal relative amplitude as a function of the storage time t_e for a coupling parameter $\alpha = 1$. The ratio T_{2b}/T_{2a} varies randomly from 5 up to 50, and M_a^0/M_b^0 from 0.05 up to 1.

$$SE(\lambda) = \|X \cdot A - S\|^2 + \lambda \|A\|^2 \quad (9)$$

where S is the measured signal and X is the exponential matrix function such that a component of the product $X \cdot A$ is $A_{ij} \exp(-t_1/T_{2i}) \exp(-t_2/T_{2j})$. The above equation is solved using matrix calculations only, involving single value decomposition (SVD) and data compression not detailed here. We focus here on the choice of the regularization parameter λ which is important for the detection of small amplitude peaks. For clarity, we illustrate the problem using a simple 1D experiment because the same procedure is used for 1D and 2D calculations. To obtain the optimum regularization value, the standard procedure is to plot the $SE(\lambda)$ curve and chose

the largest value of λ giving the lowest SE value (Fig. 5). This procedure is more important than one would think in terms of statistical significance of the small peaks. For this purpose, using a distribution containing a small peak characterizing micropores, we calculated a synthetic signal. Then we added a white noise at a ratio $S/N = 100$ and calculated a new solution according to Eq. (9). This procedure was repeated for 1000 different noise realizations. The results are plotted in Fig. 6 as a distribution band representing the highest and lowest values obtained at a given relaxation time. When the regularization parameter is not optimum and too small, the low limit of the band reaches zero amplitude for the microporosity peak, meaning that the amplitude and position of the peak are very

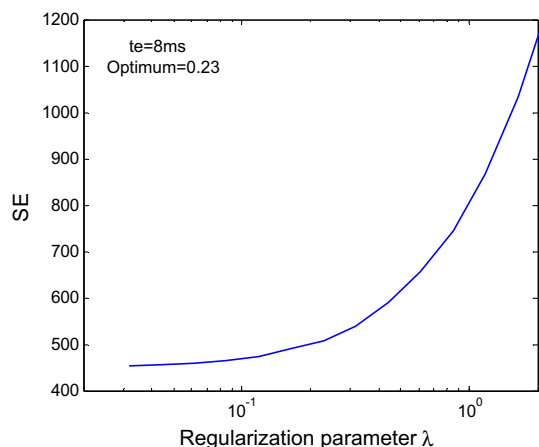


Fig. 5. Illustration of the choice of the optimum regularization parameter when computing 2D relaxation time maps. This curve corresponds to the smectite gel example described in Fig. 9 ($t_e = 8$ ms).

uncertain and depend on the noise realization. When the regularization parameter is optimum in the sense described above, the lowest limit of the error band is non-zero and one can clearly identify the micropore peak. Note that the choice of the optimum value on the $SE(\lambda)$ curve may differ slightly from one user to another because different criteria can be used to define the curve increase. The same tuning applies to the 2D analysis that uses the same algorithm and regularization principle. It is particularly important when trying to detect small amplitude peaks.

The second important aspect is the effect of the signal to noise ratio. This is illustrated on a 2D calculation in which synthetic 2D signals were generated from a known solution (Fig. 7). The square represents the position of the four peaks of amplitude $P^{--} = 0.07$, $P^{+-} = P^{-+} = 2.5$ and $P^{++} = 100$. The calculations performed for three values of signal to noise ratio indicate that small off-diagonal peaks can indeed be detected, but their location may deviate from the theoretical square position, except for the highest signal to noise ratio of 500. However, such a high value may require time consuming experiments. We usually acquire experiments at a signal to noise ratio of 300.

4. Experimental results

We studied four systems showing a bimodal pore structure in which diffusional exchange are suspected to modify the pore structure information inferred from 1D relaxation time distribution. Experiments were performed on a MARAN-ULTRA 23.7 MHz from Oxford Instruments equipped with a 18 mm proton probe. The samples are small cylinders of diameter 15 mm and length 20 mm. For the gels, we used a glass container of similar size.

4.1. Smectite gels

The first system consists of smectite clay particles that have aggregated due to inter-particle interactions occurring at high salinity (0.1 mol/L CaCl_2) and high clay concentration. A full study of diffusion properties [14] shows an exponential decrease of the effective diffusion coefficient as a function of clay fraction. When flocculated, the T_2 distribution is bimodal (Fig. 8); the lowest mode

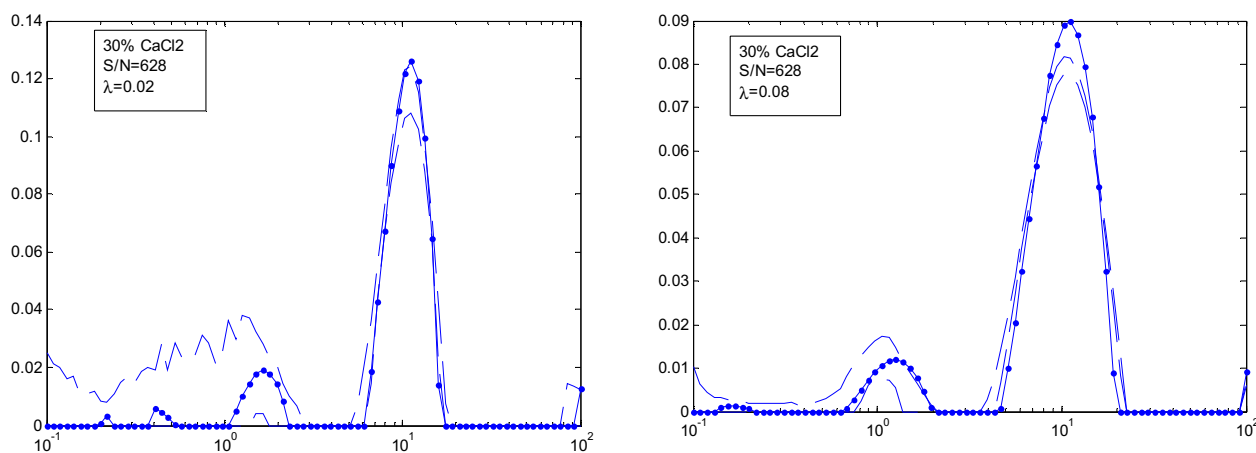


Fig. 6. Error band for two different values of the regularization parameter λ .

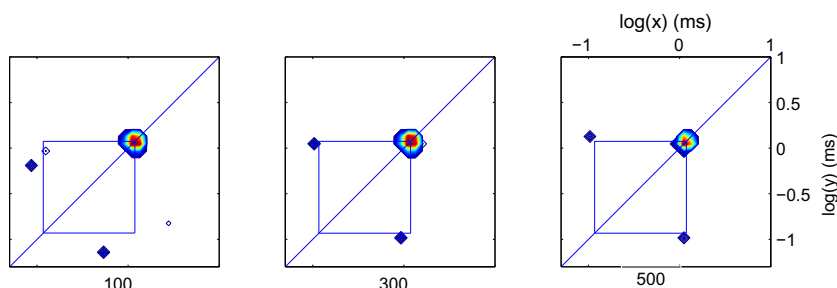


Fig. 7. Effect of the signal to noise ratio (100, 300, 500) on the detection of off-diagonal peaks in a T_2 -store- T_2 experiment. The input signal consists of four peaks of amplitude $P^{--} = 0.07$, $P^{+-} = P^{-+} = 2.5$ and $P^{++} = 100$.

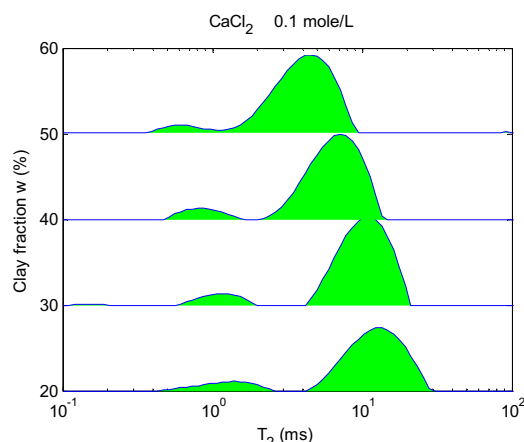


Fig. 8. Bimodal distributions obtained on flocculated smectite gels as a function of clay fraction. From Ref. [14].

corresponds to the intra-flock spacing (about 1 nm, see Fig. 8 in Ref. [14]) and the highest to the inter-flock spacing (about 10 nm), the latest varying with the amount of clay. Given the very small pore sizes in these systems, diffusional coupling may strongly affect the T_2 distribution and result in wrong conclusions about the inter- and intra-flocks fraction.

The T_2 -store- T_2 experiments were performed on the 30% clay fraction sample for increasing values of the storage time t_e , from 0.5 ms up to 10 ms. The 2D maps (Fig. 9) indicate no diagonal peaks at short exchange time ($t_e = 0.5$ ms) but they gradually appear when increasing the exchange time. Note that the amplitude color¹ coding is not the same between the different graphs, such that the amplitude decreases cannot be observed. The four peaks form a square pattern as expected but the square is not perfect for the reasons developed in the previous section. We also measured T_1 and T_2 distributions and determined that $T_1/T_2 = 1.8$.

The analysis of the peak intensities (Fig. 10) allows an estimation of the exchange rates k_a and k_b , the initial magnetization M_a^0 and M_b^0 , and the relaxation times T_{2b} and T_{2a} . We adjusted the four parameters k_a , M_a^0 , T_{2b} and T_{2a} to the curves. The intensity of the main peak is normalized by the amplitude at the smallest imposed t_e value ($P_n^{++} = P^{++}/P^{++}(t_e = 0.5 \text{ ms})$). We found $k_a = 0.06$, $M_a^0/(M_a^0 + M_b^0) = 0.155$, $T_{2b} = 11$ ms, $T_{2a} = 2$ ms. Hence, we have $k_b = 0.011$ and the degree of coupling is $\alpha = 0.28$. According to Fig. 3, this system is weakly coupled and intraflocks porosity fraction is underestimated by about 10% only.

4.2. Shaly sandstone (Vosges sandstone)

This case is typical of natural sandstones containing a variable amount of clays, either filling the pores or lining at their surface. The estimation of the pore volume fraction occupied by water in clays is of practical industrial importance because it is directly related to the potential of an oil reservoir in terms of hydrocarbon in place. Compared to the previous example, the pore sizes are changed by several orders of magnitude. Typically, clay pore sizes in such natural systems is around 1 μm or less and intergranular spacing is of order 10 μm . From a mercury injection experiment on a companion sample, the dominant throat diameter is 7 μm , and the clays are represented as a tail at smaller size extending down to 0.1 μm . However, the mercury injection experiment is somewhat ambiguous because it must be performed in a dry state.

From this point of view, the NMR T_2 characterization appears ideal because performed in a fully saturated state. However, the separation between water trapped in clays (clay bound water) and intergranular water is usually not well defined, as shown by the 1D projection of the 2D map at short exchange time in Fig. 11. It is usually considered that a cut-off around 10 ms gives reasonable estimation of clay bound water [15].

The T_2 - T_2 maps calculated at a long exchange time ($t_e = 90$ ms, Fig. 11) indicate clearly an exchange between clays and intergranular porosity (i.e. between pores at $T_2^- = 2$ ms and $T_2^+ = 50$ ms indicated by the square pattern). In a similar way as for the smectite gel, we analyzed the intensities as a function of the exchange time t_e (Fig. 12). In this case, we subdivided the T_2 maps into four regions to define off and on-diagonal intensities (Fig. 11). In this case, the intensities P^{++} , P^{--} , P^{+-} and P^{-+} are the sum of the amplitudes in each sub-region. We found $k_a = 0.02 \text{ ms}^{-1}$, $k_b = 0.016 \text{ ms}^{-1}$, $M_a^0/(M_a^0 + M_b^0) = 0.45$, $T_{2b} = 200$ ms, $T_{2a} = 10$ ms. The T_1/T_2 ratio was set at 1.8. Hence, the degree of coupling $\alpha = 0.73$ is weak to intermediate. The back calculated values of T_2^- and T_2^+ are, respectively, 8 and 55 ms. Hence, the intergranular pore size (or more exactly the V/S ratio) is underestimated by a factor of 4 on the 1D distribution.

4.3. Double porosity carbonates

Carbonates often contain multiple pore sizes due to various diagenetic processes. Contrary to sandstones, natural carbonates are made of various debris (animals, plants) that are subjected to dissolution/precipitation before, during or after burial. Hence, porosity can be present between, as well as, inside "grains". The samples used here are part of a larger study comparing NMR T_2 distributions and pore throat size distributions deduced from mercury injection experiments (MICP), with the help of visual examination of the pore structure from thin sections and scanning electron microscopy (SEM) [16,17]. In this work, it was established that microporosity defined in the petrographical sense, is the volume accessible through throat diameters smaller than 2 μm and correspond to relaxation times smaller than 200 ms in the T_2 distribution. An example is given in Fig. 13. Macroporosity is located between red algae, that are themselves porous. As a result, the MICP distribution indicates a bimodal distribution. The NMR T_2 distribution indicates similarly a bimodal distribution shifted, as expected, to the left of the MICP distribution, after calculation of the surface relaxivity from BET measurements. The macropore peak is centered around $T_2 = 1000$ ms, and we performed a 2D experiment using an exchange time of $t_e = 1000$ ms that indicated no off-diagonal peak and therefore no pore to pore diffusional exchange. The presence of sparitic cement as well as a cement rim around red algae is the most likely explanation for the lack of coupling. Also, the size of the object containing microporosity is large (300 μm) compared to the diffusion length of molecules during the exchange time (about 100 μm for 1000 ms).

A careful observation of the porous structure may give good indication of the existence of pore coupling. In the case of the Lavoux limestone (Fig. 14, left), microporosity is located in well identified spherical structures (oolithes) without cementation at their surface. In this case, the bimodal MICP signature (Fig. 14, right) is in contradiction with the 1D NMR when saturated with water (Fig. 15, left). Indeed, the MICP indicates that about 50% of the pore volume is accessible through throat diameters below 2 μm . The NMR distribution indicates a wide distribution with a small peak around 2 ms representing only 10% of the pore volume. The 2D map indicates an exchange between population located at 30 and 2 ms. Clearly, there exist very large pores (of the order of 100 μm) that should have a large relaxation time (1000 ms) similar to the previous case. Therefore, this system is strongly coupled. To

¹ For interpretation of the references to color in Fig. 9, the reader is referred to the web version of this paper.

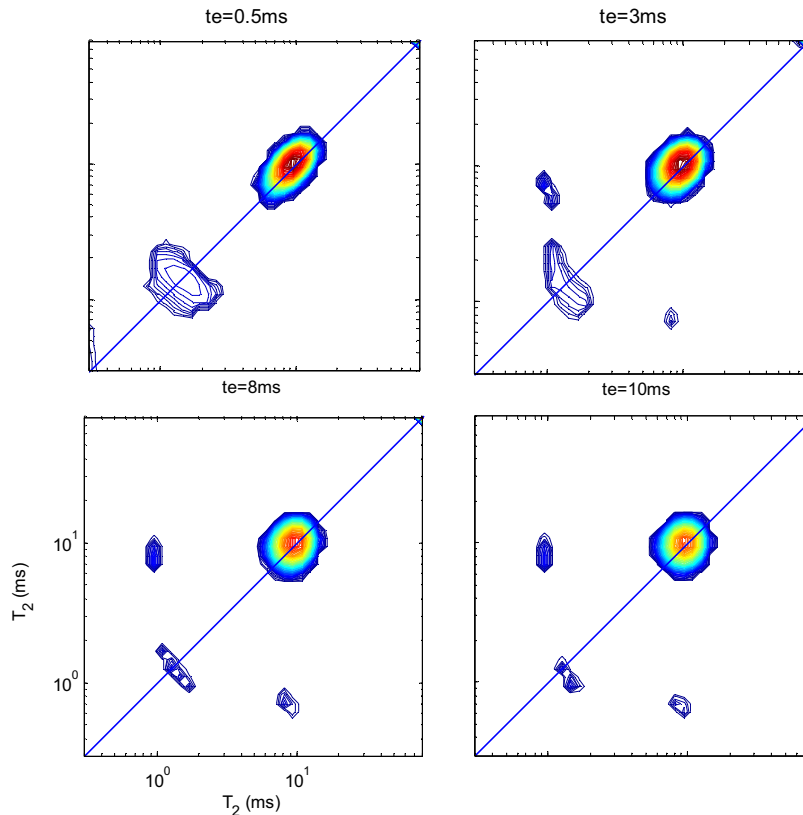


Fig. 9. T_2 -store- T_2 relaxation maps for a smectite gel at a clay fraction of 30%, as a function of the exchange time t_e . The position of the micro- and macropore peaks are independent of t_e ($T_2^+ = 9.7$ ms, $T_2^- = 1.4$ ms).

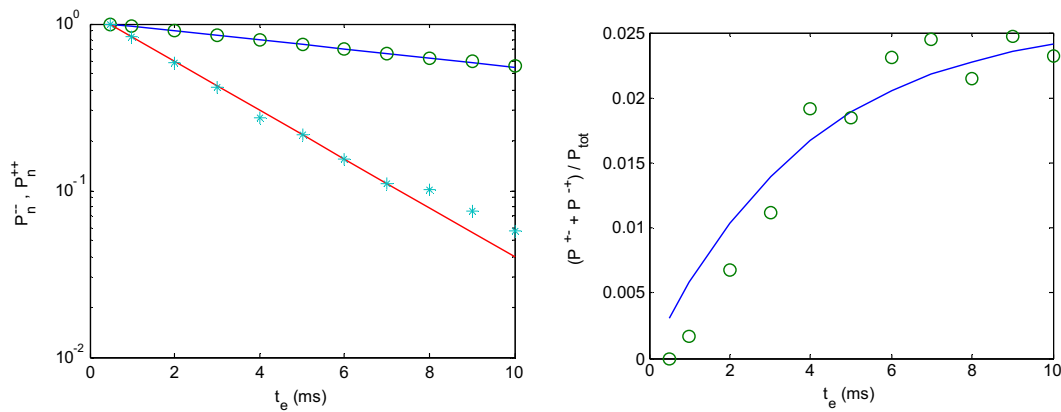


Fig. 10. Peak intensities as a function of exchange time t_e for the smectite gel at a clay fraction of 30%. Left: $P_n^{++} = P^{++}/P^{++}(t_e = 0.5$ ms) ($\circ$), $P_n^{--} = P^{--}/P^{--}(t_e = 0.5$ ms) ($*$). Right: normalized relative amplitude of the off-diagonal peak intensities. The full curves indicate a fit with $k_a = 0.060$ ms $^{-1}$, $k_b = 0.011$ ms $^{-1}$, $M_a^0/(M_a^0 + M_b^0) = 0.155$, $T_{2b} = 11$ ms, $T_{2a} = 2$ ms. The degree of coupling is $\alpha = 0.28$.

decrease the degree of coupling, the saturating fluid was replaced by a mixture of water and glycerol to a weight ratio of 36:64. Hence, the diffusivity is reduced by one order of magnitude (from 2.6×10^{-9} m 2 /s down to 0.18×10^{-9} m 2 /s at 30 °C, as measured by PFG NMR). The liquid bulk relaxation time is also reduced (from 2700 down to 380 ms) but is sufficiently large to allow the observation of large pores. In these conditions, we observed a tri modal 1D distribution more representative of the texture observed by SEM. When performing the exchange experiment using a similar exchange time (30 ms), we see two couples of diagonal peaks indicating an exchange between populations located at 30 and 2 ms, and 300 and 2 ms, but no exchange between populations at 300

and 30 ms. We interpret these observations in terms of the structure revealed by SEM visualization (Fig. 14): oolites contain microporosity but also intermediate size pores (that could be labeled mesoporosity) in their centers. Hence the exchange between meso- and macropores has a large time constant preventing the observation of off-diagonal peaks. It is also consistent with the observation of two distinct peaks in the MICP distribution: mesoporosity is only accessible through microporosity located around the oolites.

For both saturating liquids, we calculated the peak intensities in a similar way as for the Vosges sandstone and attempted to calculate the coupling coefficients and exchange rates. These attempts

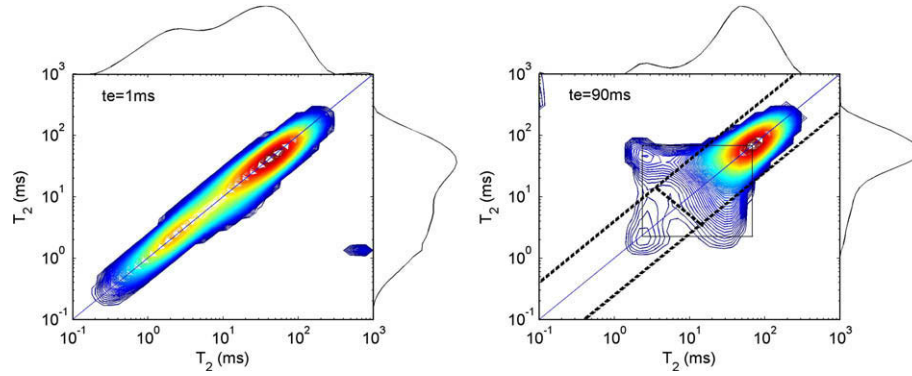


Fig. 11. T_2 -store- T_2 relaxation maps for a Vosges sandstone sample for two values of the exchange time t_e . Porosity 22% and permeability: 210 mD. The clay (micropore) and intergranular (macropore) peaks are centered around $T_2^- = 2$ ms and $T_2^+ = 50$ ms, respectively. The square pattern indicate the coupling, and the dashed lines the sub-regions for calculating peak intensities.

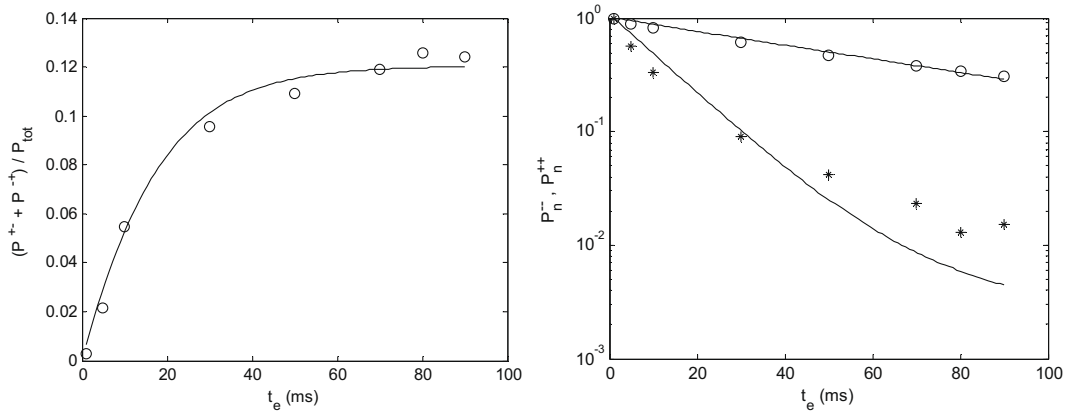


Fig. 12. Peak intensities as a function of exchange time t_e for the Vosges sandstone sample. Left: $P_n^{++} = P^{++}/P^{++}(t_e = 0.5 \text{ ms})$ ($\circ$), $P_n^{--} = P^{--}/P^{--}(t_e = 0.5 \text{ ms})$ (*). Right: normalized relative amplitude of the off-diagonal peak intensities. The full curves indicate a fit with $k_a = 0.02$, $M_a^0/(M_a^0 + M_b^0) = 0.45$, $T_{2b} = 200$ ms, $T_{2a} = 10$ ms. The degree of coupling is 0.73.

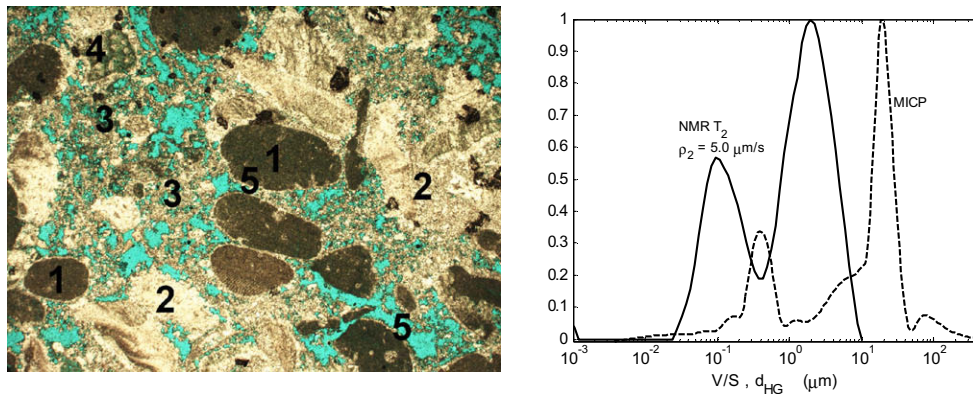


Fig. 13. Left: texture of a sample of Estailades limestone as shown by a thin section; 1: red algae; 2: sparitic cement; 3: partially dissolved cement; 4: brachiopod fragment; 5: early cement rim. Porosity 29% and permeability: 175 mD. The width of the image represents 3125 μm . The green color indicates macropores (filled with resin). Right: NMR T_2 and mercury injection distributions. The surface relaxivity ρ_2 has been calculated using specific surface measurements on the same sample (BET technique). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

were not successful and yielded inconsistent values of T_2^- and T_2^+ when compared to the observed peak center position. This not surprising because the model only takes into account two pore populations. However, we expected to be able to estimate the coupling coefficient with the lower diffusivity experiment by analyzing the

peak intensity separately. We encountered this problem in other carbonate samples having large pore size and similar complex structure (not shown). This difficulty may be linked to the existence of a distribution of exchange time. More likely, all the macropore water may not be in exchange with micropore water

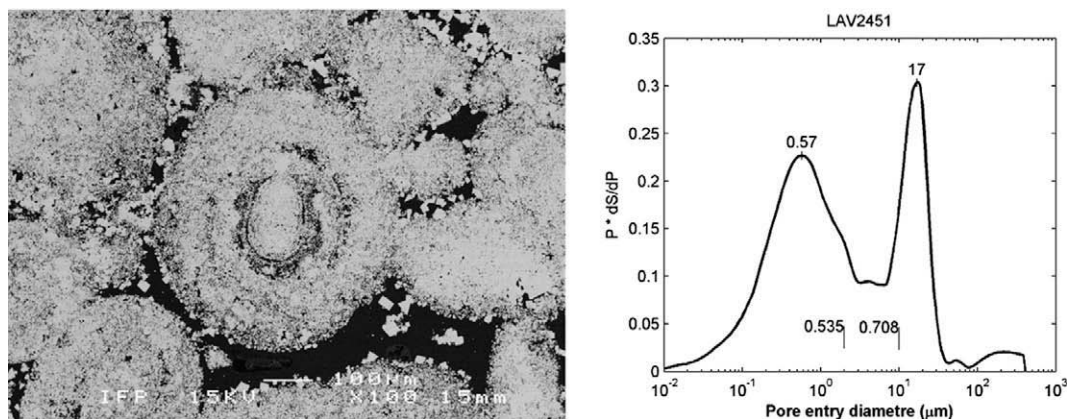


Fig. 14. Characterization of a Lavoux limestone sample. Left: SEM visualization, right: mercury injection capillary pressure curve (MICP). Porosity: 24%, permeability: 19 mD. The two pore size structure is evidenced by the bimodal MICP pore entry size distribution.

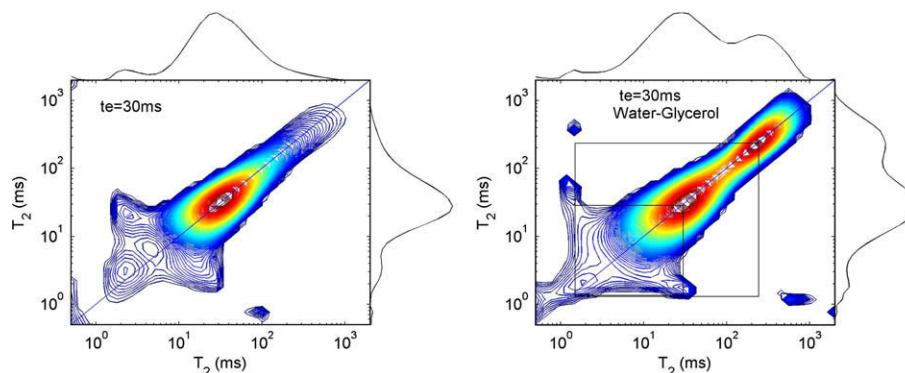


Fig. 15. T_2 -store- T_2 relaxation maps for the Lavoux limestone saturated with water (left, $D = 2.610^{-9} \text{ m}^2/\text{s}$ at 30°C) and a mixture of water and glycerol (right, $D = 0.1810^{-9} \text{ m}^2/\text{s}$, $T_{2\text{bulk}} = 380 \text{ ms}$ at 30°C). The squares highlight the exchange between micro- and mesopores, and micro- and macropores. When saturated with water, the T_2 distribution is wide and quasi-unimodal, in contradiction with the MICP distribution and the SEM visualization in Fig. 14. For a lower molecular diffusion of the saturating fluid, the pore coupling decreases and a multimodal distribution is observed.

because of the large macropore size and an insufficient diffusion length, a situation similar to the diffusion limited regime when considering surface relaxation mechanism [2]. A rough estimation can be made by comparing the experiments with water and water/glycerol: the exchange rate being proportional to diffusivity and decreased by a factor of ten, we have necessarily a coupling coefficient in the range 1–10 for water experiment, and 0.1–1 for the water/glycerol mixture experiment.

5. Conclusion

Diffusional coupling is a very common mechanism, potentially perturbing the interpretation of 1D relaxation time distributions, but providing new information of the connectivity of porous systems. For bimodal systems in which we cover a range of pore sizes from 1 nm up to 100 μm , we defined a coupling parameter α comparing the average exchange rate and the relaxation rate of the smallest pore population (micropore) having the largest specific surface. Based on an analytical solution, we analyzed the behavior of 1D T_2 and 2D T_2 -store- T_2 relaxation experiments as a function of α . In 1D experiments and for increasing values of α , the relaxation time of micropores is shifted to smaller values and their amplitude decreases. The coupling parameter, which is proportional to the diffusivity of the saturating fluid, must vary by two orders of magnitude to switch from a weakly to a strongly coupled state. For 2D

T_2 -store- T_2 experiments, diagonal peak intensities are maximum at intermediate coupling degrees and represent at maximum 10% of the total intensity for usual T_1/T_2 ratio of 1.5. Therefore, these experiments require high signal to noise ratio, of the order of 200 or larger, and a careful use of 2D inverse Laplace transform, as illustrated by various tests of the inversion process. The coupling was evidenced and quantified on a flocculated smectite gel and a shaly sandstone. Although relatively simple, the two pore population model allows the analysis of distributed bimodal systems. On complex carbonate structures, multiple exchanges were observed and explained with the help of SEM visualizations but the two pore population model could not be used for a quantitative analysis of molecular exchange. We suggest that this difficulty is either due a distribution of exchange times or the partial exchange of macropore water with microporosity in the case of large pores. An estimate of the order of magnitude of the degree is performed by changing the diffusivity of the saturating liquid by a factor of ten.

Acknowledgments

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Appendix A. T_2 -store- T_2 sequence

The T_2 -store- T_2 sequence is essentially a succession of two CPMG sequences separated by a storage time t_e :

$$P_{90} - (\tau - P_{180} - \tau - \text{echo} -)_m - P_{90} - t_e - P_{90} \\ - (\tau - P_{180} - \tau - \text{echo}_{\text{acq}} -)_n$$

All echoes intensities in the second part of the sequence are acquired, yielding a two-dimensional data set of size (m, n) such that the recorded magnetization is a two-dimensional function:

$$M(t_1, t_2) = A_{xy} \exp(-t_1/T_{2x}) \exp(-t_e/T_1) \exp(-t_3/T_{2y}), \\ t_1 = 2m\tau, \quad t_3 = 2n\tau \quad (10)$$

It is usually necessary to have small τ values to properly catch short relaxation times but this would result in a huge number of pulses when large and small relaxation times are present simultaneously such as in our carbonate systems. To limit the total number of pulses to about 10,000 (to avoid a potential heating of the sample and unstability of the power amplifier), the inter-pulse time τ is increased gradually from an initial value of 30 μs up to 100 μs (or more depending on the largest relaxation time present in the sample). Then, it stays at a constant value. During the second CPMG sequence, typically 5000 echo maximum are recorded (the value of n is tuned such as to obtain a signal decreasing to zero, as usually performed in a single CPMG sequence). Then this signal is compressed by averaging all echo maximum in 40 predefined bins having a constant log spacing in time (with origin the third P_{90}), yielding a compressed time series t_e . For the first CPMG sequence, only the m values corresponding to the 40 values of t_e are used. Hence, the two-dimensional data set acquired reduces finally to a 40×40 matrix. The advantage of this protocol is the large increase of signal to noise ratio for intermediate and large relaxation times due to compression.

The solutions described in Eqs. (5) and (6) are only valid if a $+/-z$ phase cycling is performed when averaging several signals. If this is not performed, many off-diagonal peaks may be observed, even in a single pore [11]. Adding this phase cycling to the usual CYCLOP phase cycling, a minimum of eight scans is necessary.

Appendix B

Coefficients included in Eq. (6) describing the dependence of the on- and off-diagonal peak amplitudes as a function of storage time t_e .

$$C_1 = (A_2^- - F_2) \left[A_1^- (A_2^- M_a^0 - G_2 M_b^0) - G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] \\ + (B_2^- - G_2) \left[B_1^- (B_2^- M_b^0 - F_2 M_a^0) - F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

$$C_2 = (A_2^- - F_2) \left[A_1^+ (A_2^- M_a^0 - G_2 M_b^0) + G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] (B_2^- - G_2) \\ \times \left[B_1^+ (B_2^- M_b^0 - F_2 M_a^0) + F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

$$C_3 = (A_2^+ - F_2) \left[A_1^- (A_2^+ M_a^0 - G_2 M_b^0) + G_1 (B_2^+ M_b^0 + F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^- (B_2^+ M_b^0 + F_2 M_a^0) - F_1 (A_2^+ M_a^0 + G_2 M_b^0) \right]$$

$$C_4 = (A_2^+ + F_2) \left[A_1^+ (A_2^+ M_a^0 + G_2 M_b^0) + G_1 (B_2^+ M_b^0 + F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^+ (B_2^+ M_b^0 + F_2 M_a^0) + F_1 (A_2^+ M_a^0 + G_2 M_b^0) \right]$$

$$C_5 = (A_2^+ + F_2) \left[A_1^- (A_2^+ M_a^0 - G_2 M_b^0) - G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^- (B_2^- M_b^0 - F_2 M_a^0) - F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

$$C_6 = (A_2^+ + F_2) \left[A_1^+ (A_2^+ M_a^0 - G_2 M_b^0) + G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^+ (B_2^- M_b^0 - F_2 M_a^0) + F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

where

$$A_{1,2}^{+/-} = \frac{s_{1,2}^{+/-} + R_{1,2}^a + k_a}{s_{1,2}^{+/-} - s_{1,2}^{-/+}}, \quad B_{1,2}^{+/-} = \frac{s_{1,2}^{+/-} + R_{1,2}^b + k_b}{s_{1,2}^{+/-} - s_{1,2}^{-/+}}, \\ F_{1,2} = \frac{k_a}{s_{1,2}^- - s_{1,2}^+}, \quad G_{1,2} = \frac{k_b}{s_{1,2}^- - s_{1,2}^+}.$$

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